

P1 — Certified PNT Substitution

Which Navigation Source Certifies? Alternative PNT Under a Composed Network-to-Navigation Guarantee

Research proposal, RobustUAVs.ai project · Roger Nick Anaedevha · drafted 2026-08-13
Follow-on to *From Wire to Flight* (NDSS submission)

Abstract

The parent manuscript proves that a network detector’s operating point θ bounds a certified Mission-Completion-Rate floor, and finds empirically that the decisive quantity is not the proof but the *interface*: γ , the metres of position error accrued per second of degraded navigation input. This proposal observes that γ is a property of the *navigation source*, not of the attack. Changing the positioning–navigation–timing (PNT) modality therefore moves the certified window without altering a line of the proof. We propose to measure γ per modality, to show that a cryptographically authenticated PNT source discharges the re-anchoring hypothesis the parent theorem currently assumes, and to derive from the certificate a sensor-selection rule: the minimal sensor suite whose fused γ keeps a given detector setting inside the certified window.

1 The gap

The parent manuscript certifies a GNSS-based navigation stack, and its Limitations section states the boundary plainly: Theorem 1 relies on the re-anchoring hypothesis, that at least one uncorrupted state update arrives in every certification horizon T_c . A time-delay adversary satisfies this by construction, because it postpones corrections without suppressing them. A *persistent-denial* adversary does not, and there the certified radius degrades as e^{LT_c} in the achieved denial duration.

That admission is the opening. An entire literature on GNSS-denied navigation has spent a decade building precisely the capability that restores the hypothesis, and none of it is expressed as a re-anchoring guarantee.

Two facts make the instance concrete. First, Iridium’s Satellite Time and Location signal arrives at the receiver roughly three orders of magnitude stronger than GNSS and carries cryptographic authentication, so a receiver can *reject* a spoofed or tampered navigation solution rather than fuse it [3, 1]; STL-aided inertial units are being productised for exactly this purpose [8]. Second, opportunistic navigation from low-Earth-orbit signals is no longer speculative: multi-constellation work using four Starlink, one OneWeb, two Orbcomm and one Iridium satellite reports 9.5 m 3D RMSE and 4.4 m final 3D error [2, 6], and resilient navigation under jamming has been demonstrated on signals of opportunity [5].

What nobody has asked is the composed question:

Does switching PNT source move the detector’s operating point into the certified window, and by how much?

2 The claim

Let a platform carry a set S of PNT modalities — GNSS, LEO signals of opportunity, terrain-referenced navigation, visual–inertial odometry, and inertial-only dead reckoning — each characterised by

- a **degradation rate** γ_s , in metres of position error per second of stale or withheld input;
- an **authentication state** $a_s \in \{\text{authenticated, unauthenticated}\}$; and
- an **availability process** over the mission.

Claim 1 (γ becomes a vector). *The certified window edge*

$$\theta^*(m) = \frac{m}{\gamma_{\text{eff}} e^{LT_c} H} \quad (1)$$

is evaluated against the γ of the fused estimate, so admitting a low- γ source strictly widens the window. This is arithmetic rather than new theory, and that is the point: the contribution is the measurement of γ per modality, exactly as the parent paper’s contribution was the measurement of γ for GNSS.

Claim 2 (Authentication discharges the re-anchoring hypothesis). *A cryptographically authenticated PNT source supplies an uncorrupted update per T_c even under RF denial of the unauthenticated sources, because an adversary can jam it but cannot forge it, and a jammed authenticated source is detected rather than fused. This converts the parent manuscript’s admitted limitation into a hypothesis that is satisfiable, with named hardware behind it.*

Claim 3 (The certificate induces a sensor-selection rule). *Given a corridor margin m , a detector operating point θ , and a per-modality cost, the certificate induces a partial order over sensor suites: the minimal suite whose fused γ_{eff} keeps θ inside the window. This is a procurement-grade output, and it is what an operator actually wants from a robustness certificate.*

■ *Claim 2 is the load-bearing one and also the one most exposed. Authentication prevents forgery; it does not prevent denial. The claim is therefore about the adversary’s ability to make the estimator fuse a lie, not about availability. If an adversary jams every modality at once, the hypothesis fails again and the certificate is silent. The paper must say this in the same breath as the claim.*

3 What is reused, what is new

Reused unchanged	New
Theorem 1, Lemma 1, and the re-anchoring definition	γ measured per PNT modality
certificates/engine.py	availability and authentication state in the schema
experiments/compose_certified.py	a fusion model for γ_{eff}
the SORA-anchored corridor-margin family [4]	the selection rule and its optimality argument

The schema change is small and additive: `Event.metrics` gains `pnt_source` and `pnt_authenticated`, and a `PNTModality` enum joins the definitions block. `CrossLayerPairing` is untouched, so every existing record stays valid.

4 Experiment plan

4.1 Measuring γ per modality

GNSS is done: the parent manuscript reports $\gamma = 1.195$ m/s at the receiver and 1.365 m/s after the extended Kalman filter, calibrated on three real PX4 flights [9, 7]. The remainder, in descending order of evidential strength:

1. **Inertial-only.** Measurable *today* from the same flight logs, by holding the GNSS fix and integrating the inertial measurements. This yields the dead-reckoning γ that the parent paper’s cruise discussion currently *asserts* from an error budget. Do this first: it retroactively strengthens the manuscript’s most contested paragraph, and the data is already staged.
2. **Visual-inertial.** Public VIO datasets with UAV motion profiles.
3. **LEO signals of opportunity.** The honest gap. Either a partnership for STL receiver logs, or a `physics_model` γ derived from published RMSE and update rate, tagged as such.

■ *A modelled γ must never be reported as a measured one. The parent project’s provenance discipline exists for exactly this case, and the ledger will show the tag whether or not the prose does.*

4.2 Fusion model

Begin with the conservative $\gamma_{\text{eff}} = \min_{s \in S_{\text{avail}}} \gamma_s$ over available authenticated sources, then the filter-weighted version. Report both: the conservative one is what the certificate is entitled to, the weighted one is what a deployment will actually see.

4.3 Denial sweep

Extend `experiments/sweep_full.py` with a denial-duration axis, giving the certified floor as a surface over $(\theta, \text{denial duration})$ for each modality set. The interesting reading is where the surface stops falling: that is the point at which authentication has taken over from GNSS.

4.4 Verifying the selection rule

Check the order induced by Claim 3 against brute force over all suites. The rule is only worth publishing if it agrees.

5 Relation to the parent manuscript’s limitations

Limitation	Closed?	How
The certificate assumes the filter re-anchors	fully	Claim 2 makes the hypothesis satisfiable rather than assumed
γ is a three-flight hover calibration	partly	replaces the error-budget argument with a measurement, and adds modalities; does <i>not</i> deliver the cruise-inclusive flight campaign, which remains the actual gap

6 Platform integration

A **PNT Source Selector** page: toggle modalities and their authentication state, and watch $\theta^*(m)$ and the certified floor move live with the fused γ displayed. Reuses the existing export menu, the *measured/illustrative* badge, and `/api/certify/staleness`; adds one endpoint, `/api/uav/pnt/window`.

7 Risks

- **The LEO γ may stay unmeasured.** Mitigation: the paper stands on GNSS, inertial-only and visual-inertial, with LEO as a clearly tagged modelled extension. Three measured modalities against the parent paper’s one is already the contribution.
- **The fusion model could become the contribution by accident.** Keep it conservative and unglamorous. The contribution is the measurement and the rule, not the estimator.
- **Authentication claims rest partly on vendor sources** [1]. Any security property attributed to STL must be cited to a specification or an independent evaluation before it enters a theorem statement.

8 Venue and timeline

Target *IEEE Transactions on Information Forensics and Security* or *ACM Transactions on Cyber-Physical Systems* — a measurement campaign wants journal space — or NDSS/CCS if the selection rule proves sharp. Roughly five to six months, front-loaded on the inertial-only γ , which is about a week of work on data already on disk.

References

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